Reviewer 1:

Comment 1:
Please ensure throughout that "an FCS MI algorithm" is used, as on line 85, and not "a FCS MI algorithm", as on line 84.

Response 1:
We thank the reviewer for pointing this out. The phrasing has been corrected throughout the manuscript.

Change 1:
The wording was updated in the following sections:

- Introduction: “application in **an** FCS MI framework”
- Discussion: “In this study, we developed and evaluated **an** FCS MI algorithm”

Comment 2:
It would help the reader if the boxplots corresponding to the "mars" approach were easier to distinguish in the various figures. Perhaps bolding the name of the approach would help, or making the box of the boxplot more distinct than the other ones.

Response 2:
We thank the reviewer for this helpful suggestion. We have revised the figures to make the “mars” approach more visually distinct.

Change 2:
All “mars” labels are now displayed as “★mars” in bold, improving visibility across the figures.

Comment 3:
A minor detail. In section 4.3, the author describes the computational setup, including the number of cores used for parallel computation. However, the Intel Core i7-14700 CPU only has 20 (physical) cores, but some of these cores make use of hyper-threading. Therefore, my suggestion would be to clarify that the parallel processing was performed "across 28 logical cores".

Response 3:
We thank the reviewer for this careful observation. We have clarified the description accordingly.

Change 3:
Results of imputation time:

Original text: “The analyses were conducted in R version 4.4.2, with parallel processing across 28 cores for all computations.”

Revised text: “The analyses were conducted in R version 4.4.2, with parallel processing across 28 **logical** cores for all computations.”

Reviewer 2:

Comment 1:
The manuscript needs English proofreading.

Response 1:

We thank the reviewer for this suggestion. The manuscript was reviewed by a native English-speaking colleague, and we carefully proofread and revised the text throughout to improve grammar, style, and overall readability.

Change 1:

- Original text:
  "MARS offers a balance between flexibility and computational speed, efficiently handling continuous variables without requiring the large datasets needed by tree-based models."
  Revised text:
  "MARS offers a balance between flexibility and computational efficiency, handling continuous variables efficiently without requiring the large datasets often needed by tree-based models."
- Original text:
  "With advancements in computational power, sophisticated algorithms have become essential for handling complex data structures with the aim to improve missing data imputation."
  Revised text:
  "With advances in computational power, increasingly sophisticated algorithms have become available for handling complex data structures with the aim of improving missing data imputation."
- Original text:
  "However, MI algorithms that fail to capture complex data patterns can also lead to biases"
  Revised text:
  "However, MI algorithms that fail to capture complex data patterns can also lead to biased estimates"
- Original text:
  "To introduce the MARS algorithm, we draw primarily from the explanation provided in Hastie, Tibshirani, and Friedman (2017) (pp. 321-326), which offers a more extensive summary of the method and additional theoretical background."
  Revised text:
  "To introduce the MARS algorithm, we draw primarily on the explanation provided by Hastie, Tibshirani, and Friedman (2017) (pp. 321-326), which offers a more extensive summary of the method and additional theoretical background."
- Original text:
  "The MI methods, including MARS, consistently produced estimates closer to the true parameter value, indicating their effectiveness in reducing bias that arise from missing data."
  Revised text:
  "The MI methods, including MARS, consistently produced estimates closer to the true

parameter value, indicating their effectiveness in reducing bias arising from missing data."

- Original text:
  "The degree of improvement varied across methods, with the MARS-based MI demonstrating the fastest convergence to nearly unbiased estimates, while consistently maintaining coverage around 95%."
  Revised text:
  "The degree of improvement varied across methods, with the MARS-based MI method demonstrating the fastest convergence to nearly unbiased estimates, while consistently maintaining coverage around 95%."
- Original text:
  "One aspect that likely contributed to the strong coverage performance of the MARS-based MI algorithm is its relatively higher standard errors compared to other MI methods."
  Revised text:
  "One factor that likely contributed to the strong coverage performance of the MARS-based MI algorithm is its relatively larger standard errors compared with other MI methods."
- Original text:
  "Notably, the MARS-based MI algorithm achieved the fastest convergance to satisfactory performance."
  Revised text:
  "Notably, the MARS-based MI algorithm achieved the fastest convergence to satisfactory performance."
- Original text:
  "Figure 4.2: Standard error across MI method:"
  Revised text:
  "Figure 4.2: Standard error across MI methods:"
- Original text:
  "Figure 4.3: Coverage of 95% confidence intervals across MI method:"
  Revised text:
  "Figure 4.3: Coverage of 95% confidence intervals across MI methods:"
- Original text:
  "More recently, Yongshi Deng and Lumley (2023) implemented an MI algorithm using the XGBoost technique (Chen and Guestrin 2016) within the `mixgb` `R` package, which outperformed RF and CART-based imputations."
  Revised text:
  "More recently, Yongshi Deng and Lumley (2023) implemented an MI algorithm using the XGBoost algorithm (Chen and Guestrin 2016) within the `mixgb` `R` package, which outperformed RF and CART-based imputations."
- Original text:
  "However, in many cases, the coverage for some imputation methods fell below this threshold, indicating potential issues with underestimation of uncertainty of imputing missing data."
  Revised text:
  "However, in many cases, the coverage for some imputation methods fell below this threshold, indicating potential issues with underestimation of uncertainty due to

missing data."

Comment 2:
What is the limitation of the proposed approach?

Response 2:
We thank the reviewer for this important comment. We agree that the limitations of the proposed MARS-based MI approach should be stated more explicitly. We have therefore revised the Discussion to clarify that the proposed method shares the general trade-offs of flexible, data-adaptive imputation methods: its performance may depend on the underlying data structure and on tuning choices, and increased flexibility may come at some cost in efficiency in certain settings. We also note that, because MARS is spline-based rather than tree-based, it may be particularly well suited to continuous variables. In addition, we added concrete directions for future work, including extensions to mixed-type data, multivariate missingness, and a more systematic investigation of sensitivity to tuning parameters.

Change 2:
New paragraph in discussion:
"More generally, the proposed MARS-based MI algorithm shares some of the usual limitations of flexible, data-adaptive imputation methods. In particular, its performance may depend on the structure of the data and on tuning choices, such as the allowed degree of interactions or model complexity. Thus, while the flexibility of MARS is attractive for capturing nonlinear relationships, it may in some settings come at the cost of efficiency, as reflected in the comparatively larger standard errors observed here. At the same time, because MARS uses spline-based basis expansions rather than recursive partitioning, it may be particularly advantageous for continuous variables, where smooth functional forms are plausible."